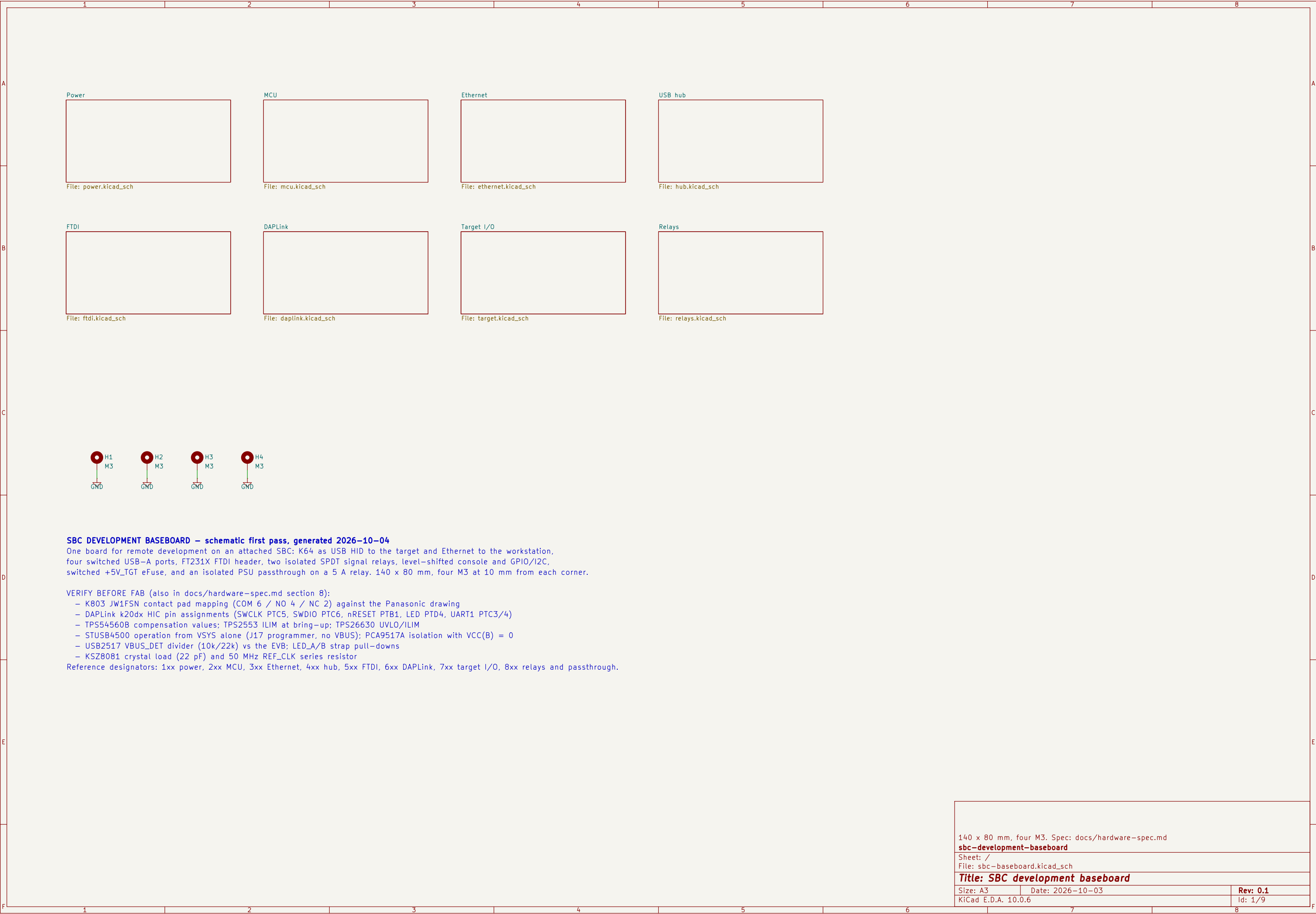




H1

M3

GND

H2

M3

GND

H3

M3

GND

H4

M3

GND

SBC DEVELOPMENT BASEBOARD – schematic first pass, generated 2026–10–04

One board for remote development on an attached SBC: K64 as USB HID to the target and Ethernet to the workstation, four switched USB–A ports, FT231X FTDI header, two isolated SPDT signal relays, level–shifted console and GPIO/I2C, switched +5V_TGT eFuse, and an isolated PSU passthrough on a 5 A relay. 140 x 80 mm, four M3 at 10 mm from each corner.

VERIFY BEFORE FAB (also in docs/hardware–spec.md section 8):

- K803 JW1FSN contact pad mapping (COM 6 / NO 4 / NC 2) against the Panasonic drawing
- DAPLink k20dx HIC pin assignments (SWCLK PTC5, SWDIO PTC6, nRESET PTB1, LED PTD4, UART1 PTC3/4)
- TPS54560B compensation values: TPS2553 ILIM at bring–up; TPS26630 UVLO/ILIM
- STUSB4500 operation from VSYS alone (J17 programmer, no VBUS); PCA9517A isolation with VCC(B) = 0
- USB2517 VBUS_DET divider (10k/22k) vs the EVB; LED_A/B strap pull–downs
- KSZ8081 crystal load (22 pF) and 50 MHz REF_CLK series resistor

Reference designators: 1xx power, 2xx MCU, 3xx Ethernet, 4xx hub, 5xx FTDI, 6xx DAPLink, 7xx target I/O, 8xx relays and passthrough.

140 x 80 mm, four M3. Spec: docs/hardware–spec.md

sbc–development–baseboard

Sheet: /

File: sbc–baseboard.kicad_sch

Title: SBC development baseboard

Size: A3

Date: 2026–10–03

Rev: 0.1

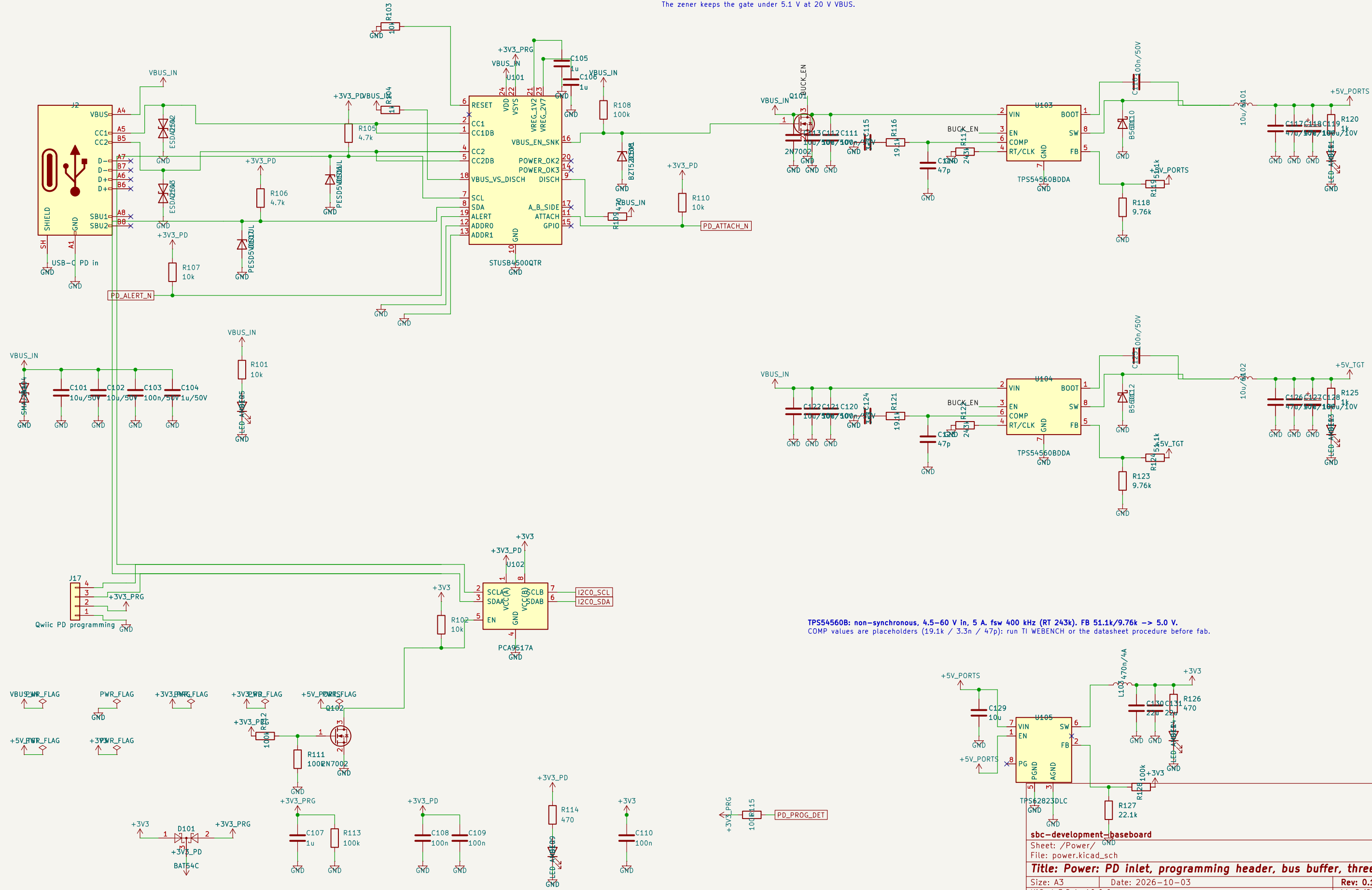
KiCad E.D.A. 10.0.6

Id: 1/9

POWER

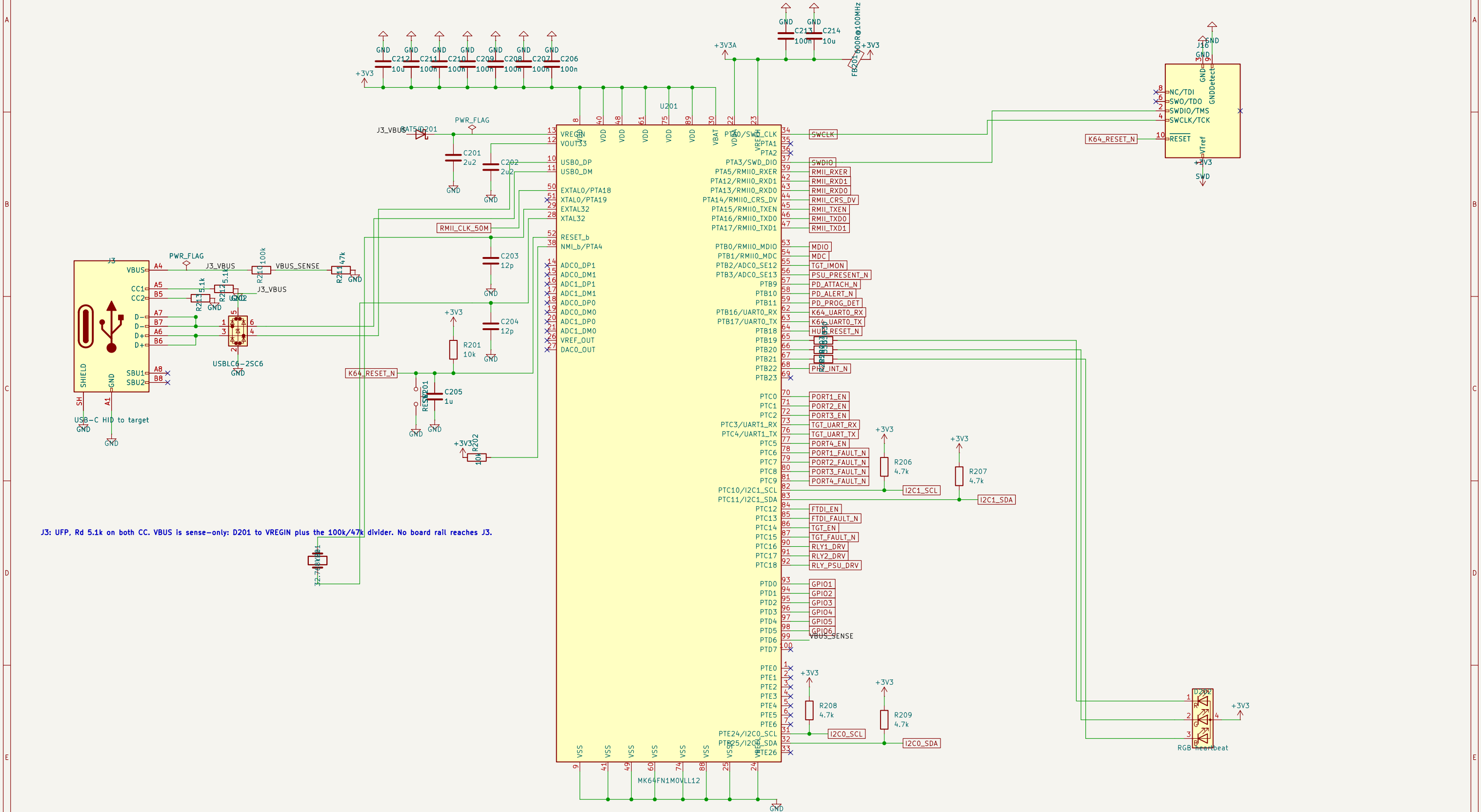
J2 is a PD sink only (no data). STUSB4500 negotiates from NVM; VBUS_EN_SNK gates the 5 V bucks through Q101, so the rails only come up once a sink contract (or plain Type-C 5 V) is valid. No series VBUS FET.
J17 (Qwiic) programs the STUSB4500 NVM and powers it through VSYS when the board is dead.
PCA9517A: A side = PD segment (+3V3_PD, diode-OR of +3V3 and the programmer), B side = K64 I2C0.
A programmer on J17 pulls PCA_EN low through Q102: the K64 is disconnected from the PD bus in hardware.

BUCK_EN: VBUS_EN_SNK released (no contract) -> Q101 on -> EN low -> bucks off.
Contract valid -> VBUS_EN_SNK low -> Q101 off -> EN floats high (internal pull-up) -> bucks on.
The zener keeps the gate under 5.1 V at 20 V VBUS.



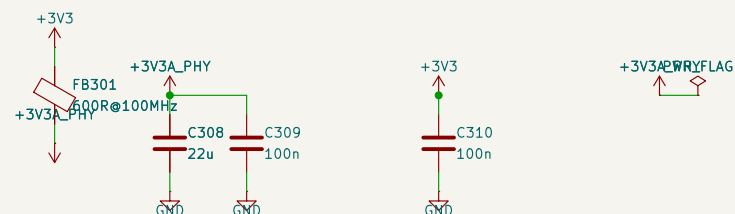
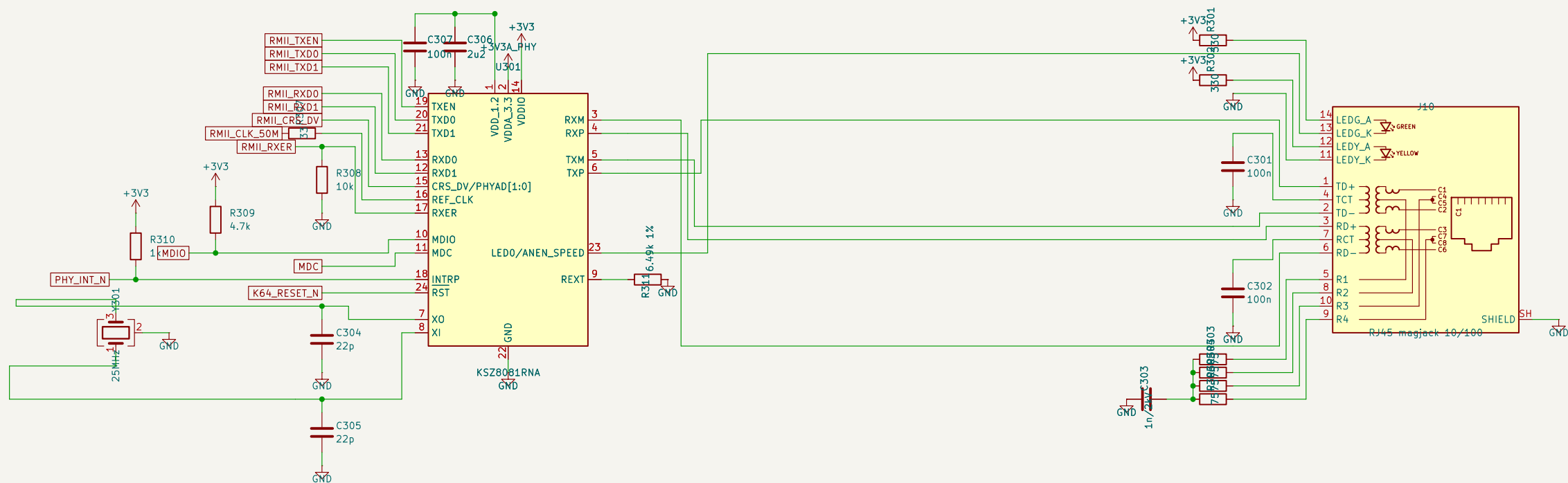
TPS54560B: non-synchronous, 4.5-60 V in, 5 A, fsw 400 kHz (RT 243k), FB 51.1k/9.76k -> 5.0 V.
COMP values are placeholders (19.1k / 3.3n / 47p): run TI WEBENCH or the datasheet procedure before fab.

MCU
MK64FN1M0VLL12 (LQFP-100). Clock: 50 MHz RMII_REF_CLK from the KSZ8081 into EXTAL0, as on the FRDM-K64F.
USB regulator: VREGIN fed only from J3 VBUS through D201, so the K64 cannot back-feed the target and the D+ pull-up disappears when the target is off. VOUT33 powers the transceiver. Pin allocation in docs/hardware-spec.md.
Spare pins (PTA1, PTA2, PTB23, PTD7, PTE0-6, PTE26, ADC, DAC) are left no-connect.



ETHERNET

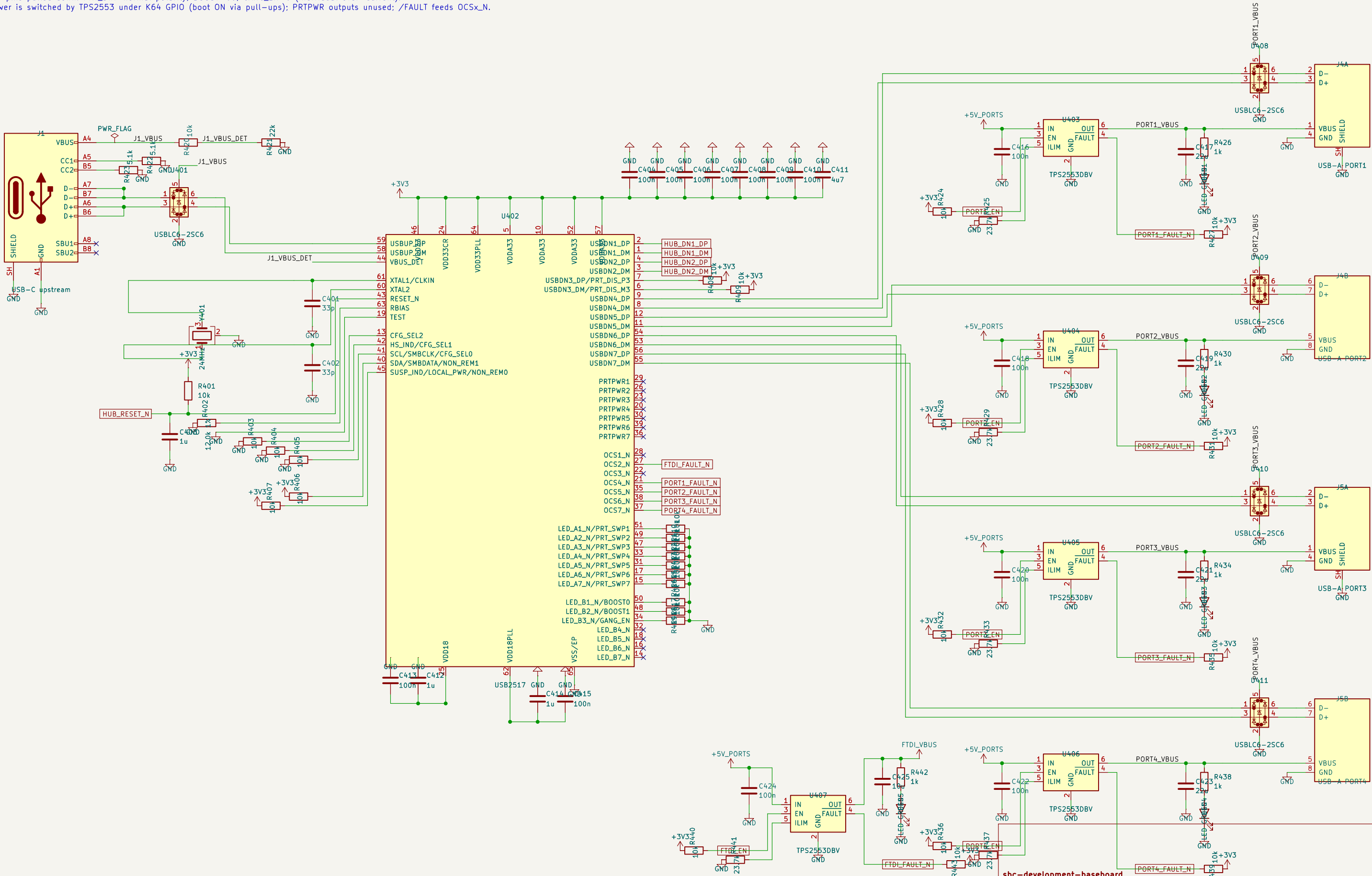
KSZ8081RNA in RMII 25 MHz mode (its power-up default): 25 MHz crystal on XI/XO, REF_CLK outputs 50 MHz to the K64 EXTALO through R301. This is the FRDM-K64F arrangement (Zephyr: microchip.interface-type = rmii-25MHz). PHY address 0 (pins 15/16 internal pull-downs; the MAC does not drive them). RXER needs the external pull-down. Transformer centre taps go to 0.1 uF each, not to a supply (datasheet 11.x). Bob Smith: 4x75R to 1 nF/2 KV.



Crystal: 25 MHz +/-50 ppm, CL 16 pF typ (datasheet) -> 22 pF each, trim at bring-up.
Green LED = link/activity (PHY LED0, active low). Yellow LED = PHY powered (cathode to GND).

USB HUB

USB2517 strap-configured (CFG_SEL=000), no firmware needed for the USB tree to come up.
Port map: 1 = DAPLink, 2 = FT231X, 3 = disabled (DN3 pulled up), 4..7 = USB-A PORT1..PORT4 (J4: ports 4,5; J5: ports 6,7).
NON_REM=11 (pin 40 up, pin 45 up) marks ports 1-3 non-removable and LOCAL_PWR high reports self-powered.
LED_A/B pins pulled down: PRT_SWP normal polarity, BOOST=00, GANG_EN=0 (individual over-current).
Port power is switched by TPS2553 under K64 GPIO (boot ON via pull-ups); PRT_PWR outputs unused; /FAULT feeds OCSx_N.

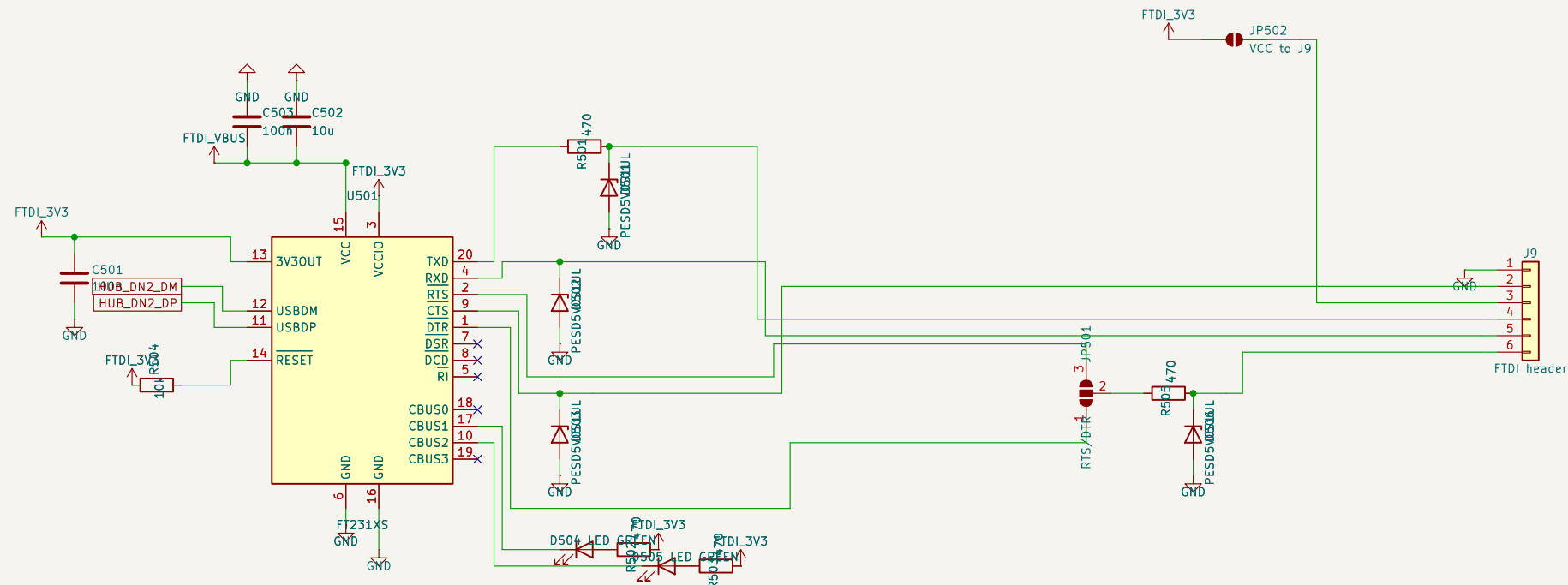


sbc-development-baseboard		
Sheet: /USB hub/		
File: hub.kicad_sch		
Title: USB hub: USB2517, upstream J1, four switched USB-A ports, FT231X s		
Size: A3	Date: 2026-10-03	Rev: 0.1
KiCad E.D.A. 10.0.6		Id: 5/9

TPS2553 ILIM 23.7k -> -1.1 A (datasheet: 20k -> 1.3 A typ, 15k -> 1.7 A). EN/FAULT pull-ups set the boot-ON state.
U407 switches the FT231X's VBUS (hub port 2) under FTDI CYCLE outside PORT ALL.

FTDI HEADER J9

Pinout = TTL-232R / SparkFun: 1 GND (blk) 2 CTS (brn) 3 VCC (red) 4 TXD (org) 5 RXD (yel) 6 RTS/DTR (grn).
TXD/RXD named from the FT231X side: TXD drives the target's RX. 3.3 V levels only; 1.8 V consoles use J13.
JP501: pin 6 = DTR# (bridge C-A, default) or RTS# (bridge C-B). JP502: pin 3 VCC from 3V3OUT (50 mA), open by default.
VCC comes from the TPS2553 on the hub sheet (FTDI_VBUS): FTDI CYCLE power-cycles the bridge without a replug.
CBUS defaults: CBUS1 = RXLED#, CBUS2 = TXLED# (factory MTP); change with FT_PROG if ever needed.



sbc-development-baseboard

Sheet: /FTDI/
File: ftdi.kicad_sch

Title: FTDI: FT231X on hub port 2, standard 6-pin FTDI header

Size: A3

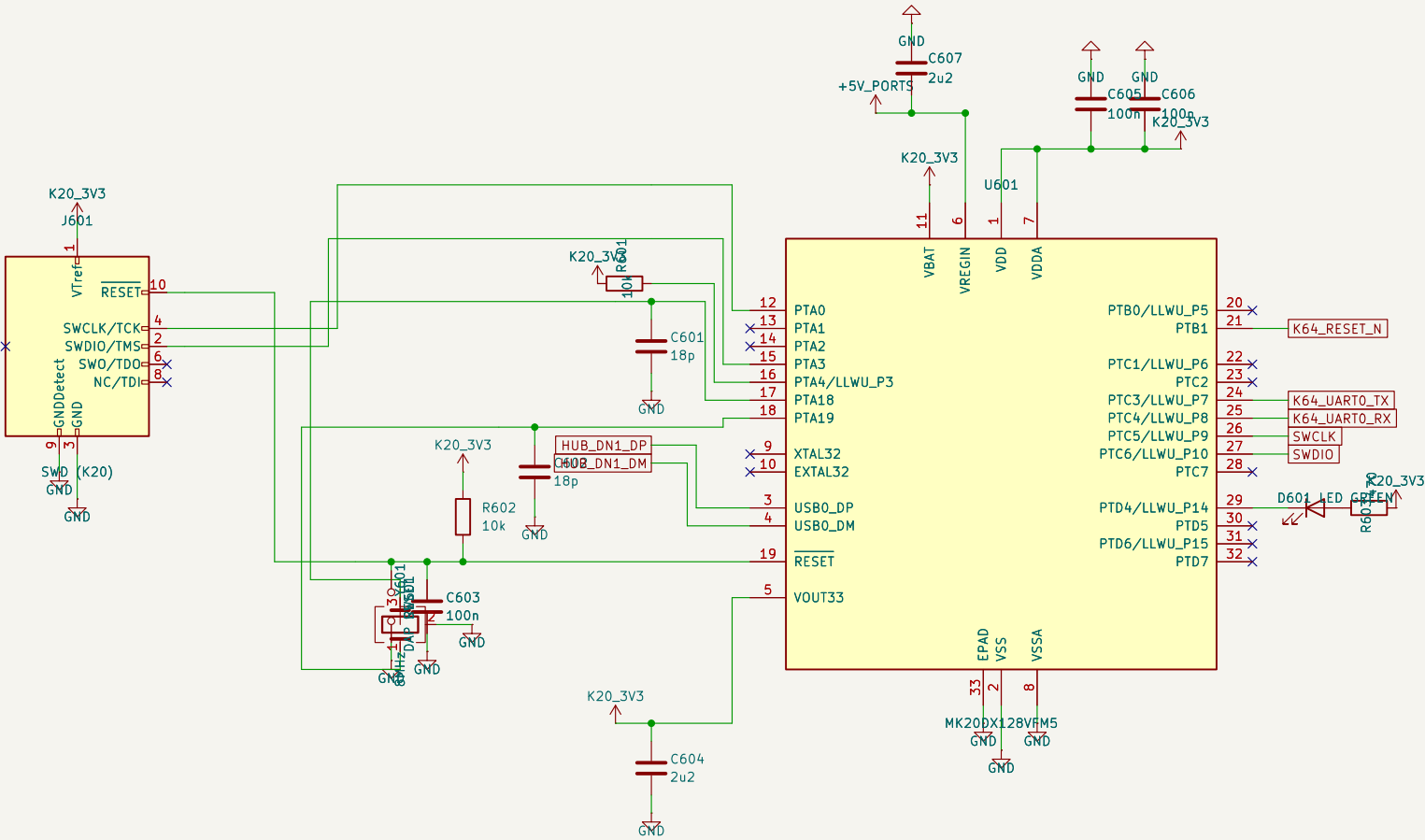
Date: 2026-10-03

Rev: 0.1

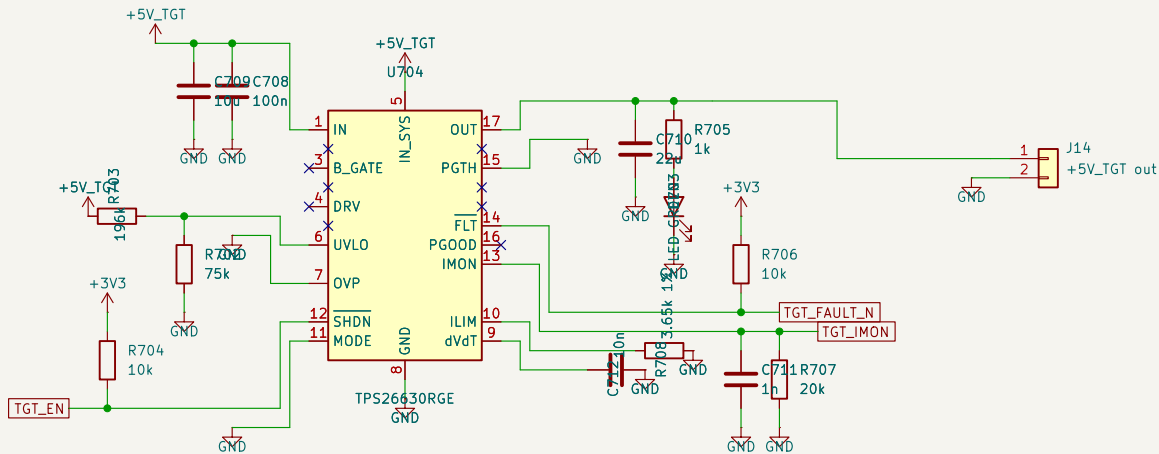
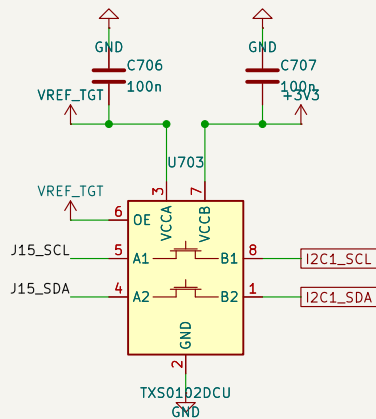
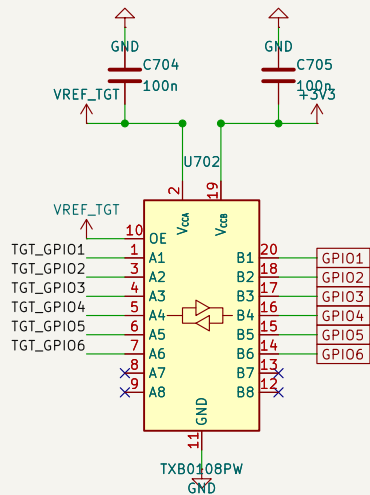
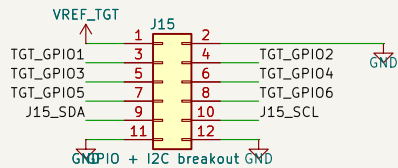
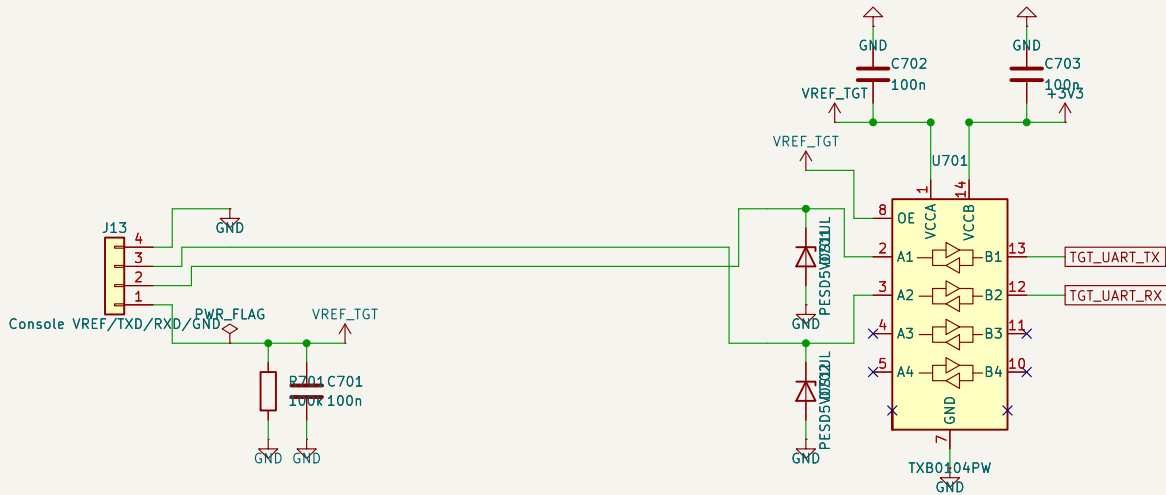
KiCad E.D.A. 10.0.6

Id: 6/9

DAPLINK
MK20DX128VFM5 running DAPLink (k20dx HIC), powered from +5V_PORTS through its own USB regulator (VOUT33 -> VDD).
Pin use follows the DAPLink k20dx HIC: PTC5 = SWCLK, PTC6 = SWDIO, PTB1 = nRESET to the K64, PTD4 = LED,
UART1 PTC3/PTC4 bridges to the K64 UART0 (CDC console). Verify against DAPLink source/hic_hal/freescale/k20dx/IO_Config.h.
J601 programs the K20 itself (bootloader + DAPLink). SW601 held at power-up enters DAPLink maintenance mode.

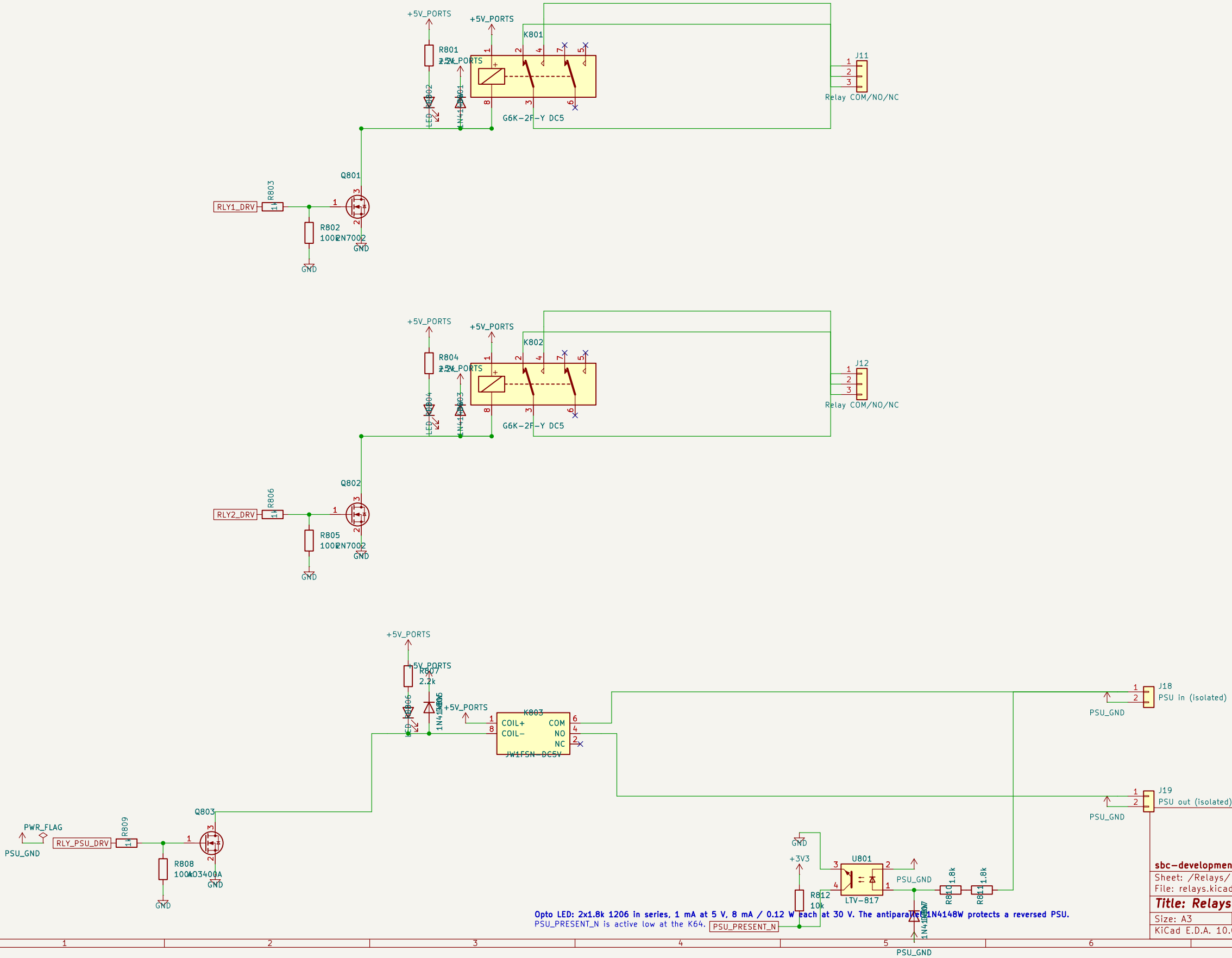


TARGET-FACING I/O
VREF_TGT comes from the target on J13.1 / J15.1 and sets the level of every translator (TXB0104/TXB0108/TXS0102).
OE pins follow VREF_TGT: with the target off the translators are high-impedance. J15 GPIO are push-pull (TXB).
TXS0102 instead of PCA9306: works with VREF_TGT equal to 3.3 V (PCA9306 needs VREF2 > VREF1). Range 1.65–3.6 V.
TPS26630: ILIM 3.65k → -5 A, UVLO 196k/75k → 4.3 V, OVP/MODE/PGTH to GND (latch-off). IMON 20k → 2.8 V at 5 A into PTB2.



RELAYS AND PASSTHROUGH

K801/K802 G6K-2F-Y: pole 1 used (COM 3, NC 2, NO 4), pole 2 unused. Coils boot de-energized (gate pull-downs).
Red LEDs sit across each coil and show the energized state. Flyback 1N4148W on every coil.
Passthrough J18 -> K803 (NO contact) -> J19 on nets PSU_VP / PSU_VOUT / PSU_GND that touch no board net: PSU_GND is its own ground.
Only the optocoupler's emitter sits across the passthrough, sized for 5-30 V. No indicator LED there.
K803 JW1FSN pad map COM=6 NO=4 NC=2: VERIFY against the Panasonic terminal drawing before fab.



sbc-development-baseboard

Sheet: /Relays/
File: relays.kicad_sch

Title: Relays: two SPDT signal relays, isolated PSU passthrough

Size: A3	Date: 2026-10-03	Rev: 0.1
KiCad E.D.A. 10.0.6		Id: 9/9